INCH-POUND

MIL-E-82901(OS)
4 January 1995
SUPERSEDING
WS 22347A
4 September 1990

MILITARY SPECIFICATION

EXPLOSIVE, PLASTIC-BONDED, CAST PBXN-110

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two types of cast PBXN-110 plastic-bonded explosive (see 6.1).

1.2 Classification. The plastic-bonded explosive covered by this specification shall be of the following types:

Type I	Polymethylene polyphenylisocyanate cured
Type II	Isophorone diisocyanate cured

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-L-3061	Lecithin
MIL-H-45444	HMX

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1376

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-650	Explosive: Sampling, Inspection and Testing
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of military specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

OTHER GOVERNMENT DOCUMENTS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

AS 2328	Isodecyl Pelargonate

NAVAL SEA SYSTEMS COMMAND (CAGE Codes 10001 and 53711)

OS 9804	Ferric Acetylacetonate
WS 14418	Dibutyltin Dilaurate
WS 16305	Isophorone Diisocyanate
WS 16321	4, 4'-Methylene Bis (2,6-Di-Tert-Butylphenol)
WS 23148	Polybutadiene, Liquid, Hydroxyl-Terminated
WS 22348	Polymethylene Polyphenylisocyanate
WS 22349	Dibutyltin Sulfide

(Application for copies should be addressed to the Commander, Naval Surface Warfare Center, Indian Head Division (Code 8410P), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

CHEMICAL PROPULSION INFORMATION AGENCY

CPIA Publication No. 21 JANNAF Solid Propellant Mechanical Behavior Manual

(Application for copies should be addressed to the Chemical Propulsion Information Agency, Applied Physics Laboratory, Johns Hopkins University, Johns Hopkins Road, Laurel, MD 20810.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Formulation (M101). The PBXN-110 shall be a plastic-bonded, castable material that uses an elastomeric polymer to incorporate solid powders and thereby produce a flexible, cured explosive. The formulation of explosive PBXN-110 shall comply with the requirements specified in table I (see 6.7).

3.2.1 Cure conditions. The explosive shall be cured at a recommended temperature of between 20 and 50°C until the Shore A hardness meets the requirements of 3.3.

3.2.2 Equivalent ratio. The weight percent of the polybutadiene and the isodecyl pelargonate shall be kept in a ratio of 1:1 $\pm$ 0.05. The weight percent of the polybutadiene and the isocyanate shall be calculated using the following NCO/OH ratio:

Type I — 0.75/1.0 to 0.85/1.0
Type II — 1.0/1.0 to 1.10/1.0

The NCO/OH ratio shall be calculated using the equivalent weights of materials obtained by chemical analysis using the techniques of 4.6.1 of WS 22348, 4.5.2 of WS 16305, and 4.6.2 of WS 23148. To avoid shelf life problems with component materials, this chemical analysis must be conducted at least every six months on lots of materials being used. The weight percent of each ingredient shall be determined to two significant figures (see 6.4).

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TABLE I. PBXN-110 formulation.

Ingredient	Specification	Percent by Weight	
		Type I	Type II
HMX, Grade B, Class 3	MIL-H-45444	[1]66.0	[1]66.0
HMX, Grade B, Class 2	MIL-H-45444	[1]22.0	[1]22.0
Polybutadiene, Linear, Hydroxyl-Terminated	WS 23148	[2]5.378	[2]5.365
Isodecyl Pelargonate	AS 2328	[2]5.378	[2]5.365
4, 4'-Methylene Bis (2,6-Di-Tert-Butylphenol)	WS 16321	0.05 ± 0.005	0.05 ± 0.005
Lecithin	MIL-L-3061	0.70 ± 0.05	0.70 ± 0.05
Polymethylene Polyphenylisocyanate	WS 22348 Type I	[2]0.494	–
Isophorone Diisocyanate	WS 16305	–	[2]0.510
Dibutyltin Dilaurate (DBTDL)	WS 14418	–	[3]0.01
Ferric Acetylacetonate (FeAA)	OS 9804	–	[3]0.0015
Dibutyltin Sulfide (DBTS)	WS 22349	–	[3]0.10

[1] The weight ratio of the HMX Class 3 to HMX Class 2 is nominally 3:1. To accommodate lot-to-lot variations in the HMX classes (see 6.5), this weight percent ratio is allowed to vary from a minimum of 1:1 to a maximum of 8.78:1. These ratios will result in the following minimum and maximum weight percentages:

Ingredient	Min.	Max.
HMX Class 3	44.0	79.0
HMX Class 2	9.0	44.0

in such a way that the total HMX content remains between 86 percent and 89 percent (see 3.3).

[2] These are nominal values assigned to these component materials. The nominal composition of PBXN-110, Type I and Type II, was derived using the values of the sample calculations (6.4.1) and the requirements of 3.2.2. Actual weight percents of these materials must be calculated in accordance with 3.2.2 using experimentally determined equivalent weights.

[3] Use either DBTDL, FeAA, or DBTS. The tolerances on the weight percent of the catalysts shall be:

Catalyst	Min.	Max.
DBTDL	0.002	0.02
FeAA	0.001	0.02
DBTS	0.03	0.30

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3.3 Cured explosive requirements. The cured explosive shall meet the requirements of table II when tested in accordance with section 4 (see 6.6).

TABLE II. Cured explosive requirements.

Property	Type I		Type II		Test Method
	Min.	Max.	Min.	Max.	
Density at 25°C (g/cc)	1.62	1.70	1.62	1.70	4.6.1
Stress, max., at 25°C (psi)	20	—	20	—	4.6.2
Strain, max. stress at 25°C (%)	10	—	9	—	4.6.2
Hardness, Shore A, 15 second delay at 25°C	15	—	20	—	4.6.3
Vacuum Stability at 100°C (mL gas per g per 48 hours)	—	0.4	—	0.4	4.6.4
Average composition, % HMX	86	89	86	89	4.6.5

3.4 Workmanship. The explosive material shall be free from foreign materials and manufactured in a manner to assure compliance with this specification. The sample shall then be evaluated to determine acceptability and the approved standards of workmanship will thereby become a minimum requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Classification of characteristics. The characteristics verified by the inspections herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Inspections that verify critical characteristics are identified by the symbol (C) and inspections verifying major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the inspection. Inspections which are not annotated with a classification code and serial number are classified minor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable inspection paragraph.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on samples which have been produced with equipment and procedures normally used in production. Any production prior to acceptance of the first article sample shall be at the contractor's risk.

4.4.1 First article sample. The sample shall consist of two sets of representative explosive material selected from a minimum batch of 225 kilograms after isocyanate has been added. A set shall contain approximately 2000 g of explosive. A set shall be composed of two specimens of explosive, one selected from the first third of the batch and another separate specimen selected from the final third of the batch. A batch is defined as that quantity of material that has been subjected to one or more chemical or physical processes (or combination thereof) intended to produce a desired product having substantially uniform characteristics.

4.4.2 Inspection routine. Material from one of the two sample sets from the batch shall be subjected to the tests specified in 4.6. The second sample set shall be forwarded to the procuring activity in a form suitable for the tests specified in 4.6. Failure to meet the requirements of this specification shall be cause for rejection of the first article sample.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.6. The quality conformance sample shall be the same as that of 4.4.1 except that it shall consist of one set. Uncured material for analysis shall be selected from the same containers from which the cast samples are poured. Explosive failing to meet any of the requirements of 3.3 when tested in accordance with 4.6 shall be classified defective, resulting in the rejection of the batch from which it was obtained.

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4.5.1 Sample identification. Each explosive sample container and specimen shall be marked with the following information:

a. Complete explosive designation
b. Lot number
c. Weight of the lot
d. Manufacturer's name and plant designation
e. Contract number
f. Date of manufacture and sampling

4.6 Test methods. All explosive specimens used in the tests of 4.6.1 through 4.6.5.1 shall be cured to a constant Shore A hardness meeting the requirements of 3.2.1.

4.6.1 Density.

4.6.1.1 Samples. Density shall be determined on two samples cut from two separate cured test specimens (one sample taken from the beginning of the pour and one sample taken from the end of the pour). Each sample should weigh between 10 and 20 g.

4.6.1.2 Density determination. The density of the sample shall be determined on an analytical balance equipped for underwater weighing. The sample shall be weighed to an accuracy of 0.1 milligram, first in air then in water.

4.6.1.3 Density calculation. The density shall be calculated according to the following formula:

$$D = \frac{W_a}{W_a - W_b} \times D_w$$

Where:
D = Density of test sample, g/cc
D_w = Density of water at test temperature, g/cc
W_a = Weight of test sample in air, g
W_b = Weight of test sample in water, g

4.6.1.4 Failure. If a sample selected for density fails to meet the requirements of table II, another sample shall be taken from the same section of mix and tested. If the second sample meets the density requirement, the batch shall be considered to meet the density requirements of this specification.

4.6.2 Stress and strain at maximum stress. Stress and strain at maximum stress shall be determined in accordance with JANNAF Solid Propellant Mechanical Behavior Manual, CPIA Publication No. 21, Section 4.3.2, Uniaxial Tensile Tests at Constant Strain Rate, Nov. 1970. Class C testing shall be used except final specimens may be die cut from the half-inch slices from the bulk sample. The specimens for testing may be selected from the first third of the batch or the final third when sampled as described in 4.4.1.

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4.6.3 Shore A hardness test (15-second delay).

4.6.3.1 Sample. The explosive of 4.5.1 in a suitable form as in 4.6.1.1 shall be tested for Shore A hardness. Any sample which exhibits bleeding at the surface at the end of the cure period shall not be tested.

4.6.3.2 Shore A hardness determination. The hardness of the cured material shall be determined using a Type A-2 durometer together with a constant load operating stand (Code DRCL). The total weight of dead weight, holding screw, and durometer shall be 1065 g. Test specimen shall be placed under the durometer pointer with the pointer resting on the surface of the sample. Release the lock called a "Pawl." Do not permit the point to shock the sample. Record the instrument dial reading 15 seconds after release. Use a stopwatch to measure the time. Determine three values each on the top and bottom of the specimen. Report the average of the top average and the bottom average as the test value.

4.6.4 Vacuum stability test. The vacuum stability test procedure shall be as described in MIL-STD-650, Method 503.1.1 except sample size shall be at least 0.2 g. Sample shall be cut into small cubes with a maximum size of 3 millimeters before being placed in the vacuum stability test tube. Two samples, one from each section of the batch, shall be tested.

4.6.5 HMX analysis.

4.6.5.1 Detailed analytical procedure, cured PBXN-110. Weighed samples of cured material shall be extracted with carbon tetrachloride (CCl_4) and the residue dried and weighed (loss in weight = CCl_4 extractables). The residue shall then be extracted with hot dimethylsulfoxide (DMSO), dried and weighed (loss in weight = RDX, HMX). The remaining residue weight is the amount of binder or polymer.

a. Cut sample into thin strips. Weigh 2.0 to 2.5 g of the thin strips to the nearest 0.1 mg into a 100-mL beaker.

b. Add 40 mL of CCl_4 and continuously stir the solution for 15 minutes.

c. Filter the solution through a weighed, medium porosity, sintered-glass crucible and quantitatively transfer the residue to the crucible.

d. Dry the crucible plus residue at 100°C to a constant weight.

e. Quantitatively transfer the residue to a 100-mL beaker.

f. Add 30 mL of hot DMSO (60 to 70°C). Heat the solution for 15 minutes with constant stirring.

g. Filter off the DMSO through a previously weighed glass crucible.

h. Repeat step f.

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i. Repeat step g.

j. Repeat step f (total volume of hot DMSO = 90 mL). Mash the sample residue with a flat-tipped stirring rod to release any trapped HMX-DMSO from the polymer matrix.

k. Filter quantitatively solid and solution through the previously weighed crucible. Mash residue with a flat-tipped stirring rod.

l. Wash residue at least 10 times with 20 mL of water and dry at 100°C to a constant weight.

Calculations:

$$\%\ CCl_4\ extractables = \frac{A-B}{A} \times 100$$

$$\%\ HMX,\ RDX = \frac{B-C}{A} \times 100$$

$$\%\ Binder = \frac{C}{A} \times 100$$

Where:
A = Initial weight of PBXN-110
B = Weight of residue after CCl_4 extraction
C = Weight of residue after DMSO extraction, polymer, or binder weight

4.6.5.2 Alternate test procedure, uncured PBXN-110.

a. Determine the HMX content of uncured PBXN-110 in duplicate on each of the two test specimens from a set of 4.5.1. Weigh a 2-gram sample of uncured explosive to an accuracy of 0.1 mg on an analytical balance prior to beginning polymerization. Place the weighed sample into a tared, 30-mL, fine porosity, sintered-glass filtering crucible. A glass stirring rod is to be included in the tare weight of the crucible.

b. Extract the sample at ambient temperature using a vacuum filtrator with five portions of 10 mL each of reagent grade methylene chloride saturated with HMX. The extractions should have a stirring time of approximately 1 minute each with the vacuum off while stirring.

c. Dry the sample for 30 minutes at 90 to 100°C, cool, and weigh.

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d. Calculate percent HMX:

$$\% \ HMX = \frac{A-B}{C} \times 100$$

Where:
A = Weight of crucible plus residue, g
B = Weight of crucible, g
C = Weight of sample, g

4.6.6 Inspection of packaging. Packaging shall be inspected as specified in section 5.

5. PACKAGING

5.1 Packaging. The explosive material shall be packaged in accordance with the manufacturer's best commercial practice.

5.2 Packing. The material, packaged as specified in 5.1, shall be packed to ensure carrier acceptance and safe delivery to the destination at the lowest applicable rate.

5.3 Special marking. In addition to any special markings required by the contract or purchase order (see 6.2), all container markings shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended Use. PBXN-110 is a high-energy explosive intended for use in ordnance. PBXN-110, Type II is the preferred composition for general use. The material has been formulated to provide a longer pot life, i.e., time between the end of mixing and the time when the material can no longer be cast into a warhead to give a high-quality charge because of its high viscosity increase. PBXN-110, Type I has a shorter pot life, but is equivalent to Type II with respect to sensitivity and performance properties. The material can be used in place of Type II for applications not requiring a long pot life, if isophorone diisocyanate is not available.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether first article inspection is required (see 3.1).

d. Testing activities to perform tests, if other than as specified (see 4.1).

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e. Any special marking required (see 5.3).

f. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.4.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Equivalent ratio. The following method is used to calculate the weight percent of polybutadiene, isodecyl pelargonate, and isocyanate:

Given: The hydroxyl value of the polybutadiene = A, meq/g
The weight of equivalent of the polybutadiene = 1000/A, g/eq
The weight of equivalent of the isocyanate = B, g/eq

Step 1: Determine the weight in grams of isocyanate required to react with each gram of polybutadiene. This is accomplished using the following formula:

$$X = \frac{BR}{1000/A}$$

Where:
X = Weight in grams of isocyanate per gram of polybutadiene
R = Mole ratio of isocyanate to polybutadiene

Step 2: Since the polybutadiene and the isodecyl pelargonate are in a 1:1 ratio, the weight percents of polybutadiene, isodecyl pelargonate, and isocyanate are in a 1:1:x ratio. The weight percent of isocyanate is equal to (X/(2 + X)) x 100, and likewise, the weight percent of polybutadiene and isodecyl pelargonate would be equal to 1/(2 + X) x 100, in the binder.

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Step 3: Since 11.25 percent of the Type I and 11.24 percent of the Type II explosive composition is made of these three ingredients , then:

1. Weight percent of polybutadiene in the two types of explosives is as follows:

$$\textit{Polybutadiene in Type I explosive} = \frac{100}{(2 + X)} (0.1125)$$

$$\textit{Polybutadiene in Type II explosive} = \frac{100}{(2 + X)} (0.1124)$$

2. Weight percent of isodecyl pelargonate in the two types of explosives is as follows:

$$\textit{Isodecyl pelargonate in Type I explosive} = \frac{100}{(2 + X)} (0.1125)$$

$$\textit{Isodecyl pelargonate in Type II explosive} = \frac{100}{(2 + X)} (0.1124)$$

3. Weight percent of isocyanate in the two types of explosives is as follows:

$$\textit{Isocyanate in Type I explosive} = \frac{100X}{(2 + X)} (0.1125)$$

$$\textit{Isocyanate in Type II explosive} = \frac{100X}{(2 + X)} (0.1124)$$

6.4.1 Sample calculation for PBXN-110.

6.4.1.1 Type I.

Given:
The hydroxyl value of the polybutadiene = A = 0.84 meq/g
The weight equivalent of the polybutadiene 1000/hydroxyl value = 1190.48 g/eq
The weight equivalent of the isocyanate = B = 137.4 g/eq
NCO/OH ratio = R = 0.80

Using the values given above and the procedures outlined in 6.4, the following is computed:

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$$X = \frac{(137.4)(0.8)}{1190.48} = 0.92 \text{ g isocyanate/g polybutadiene}$$

The weight percent of the polybutadiene is calculated using Step 3-1 of 6.4 as follows:

$$\frac{100}{2 + 0.092} (0.1125) = 5.378\% \text{ polybutadiene}$$

And likewise, the weight percent of the isodecyl pelargonate using Step 3-2 of 6.4 is:

$$\frac{100}{2 + 0.092} (0.1125) = 5.378\% \text{ isodecyl pelargonate}$$

Finally, the weight percent of isocyanate is determined using Step 3-3 of 6.4 giving the following:

$$\frac{(100)\ (0.092)}{2 + 0.092} (0.1125) = 0.495\%$$

6.4.1.2 Type II.

Given:

The hydroxyl value of the polybutadiene = A = 0.84 meq/g
The weight equivalent of the polybutadiene 1000/hydroxyl value = 1190.48 g/eq
The weight equivalent of the isocyanate = B = 113.65 g/eq
NCO to OH ratio = R = 1.0

Using the values given above and the procedures outlined in 6.4, the following is computed:

$$X = \frac{(113.65)\ (1.0)}{1190.48} = 0.095 \text{ g isocyanate/g polybutadiene}$$

The weight percent of the polybutadiene is now (Step 3-1):

$$\frac{100}{2 + 0.095} (0.1124) = 5.365\% \text{ polybutadiene}$$

And likewise the weight percent of the isodecyl pelargonate is (Step 3-2):

$$\frac{100}{(2 + 0.095)} (0.1124) = 5.365\% \text{ isodecyl pelargonate}$$

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And the weight percent isocyanate is (Step 3-3):

$$\frac{(100)\ (0.095)}{(2\ +\ 0.095)}\ (0.1124)\ =\ 0.510\%\ \textit{isocyanate}$$

6.5 HMX batches. HMX batches as received tend to vary considerably from batch to batch. It is therefore recommended that a 1-gallon (6-kg), or smaller, batch of PBXN-110 be processed and the initial mix viscosity determined for all new batches of HMX. If the initial viscosity is excessive (greater than 10 kp) thus reducing the mix pot life, then the following techniques may be tried in order to reduce the mix viscosity and/or increase pot life:

a. Shift the HMX3/HMX2 ratio from 3:1 to a ratio containing more or less Class 2 HMX.

b. To maximize pot life, the mix temperature should be kept at 25°C or less. In addition, the catalyst level can be reduced to a lower level, but one which is compatible with the limits established in 3.3.

c. Reduce the total HMX content to 87 percent.

6.6 Moisture content. To obtain proper physical properties in the cured explosive, the moisture content of the component raw materials and the moisture content of the uncured explosive should be less than 0.05 percent. The moisture content can be determined by using ASTM D 1744. In order to meet the stress and strain property requirements of 3.3, these property measurements should be made soon after meeting the Shore A hardness requirements of 3.3.

6.7 Materials. Materials which have been used to formulate PBXN-110 satisfactorily are available from the following sources.

a. Polybutadiene: available from Elf Atochem as Poly BD R45M or R45HT.

b. Isodecyl pelargonate: available from Emerdy Industries as Emery 2911.

c. 4, 4′-Methylene Bis (2, 6-Di-Tert-butylphenol): available from Ethyl Corporation as Ethyl 702.

d. Isophorone diisocyante: available from Huls America as IPDI.

e. HMX Class II and III: available from Holston Defense Corporation.

f. Dubutyltin dilaurate (DBTDL): available from CasChem, Inc. as COTIN 200 Catalyst.

g. Ferric Acetylacetonate (FeAA): available from Eastman-Kodak Corporation as Kodak #8731.

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6.8 Subject term (key word) listing.

Chemical analysis
Elastomeric polymer
Ordnance

Preparing Activity:
Navy - OS
(Project 1377-N479)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER	(1) Commercial 301-743-4358/4510	(2) AUTOVON 354-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVE., (CODE 8420) INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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